Part 5 Brief · CI eval gate and live evals

mcp-ops-server · 18 Oct 2026 · PR #6 · Status: built and green; first live run waits on your API key and model ID

WHAT WAS BUILT

- **An eval gate on every push and PR:** a dedicated CI job runs all 31 golden tasks against the real server and all 12 negative controls. Free (no model, no secrets); any failure fails the build. The report lands in the job summary and as an artifact.
- **A live OpenAI agent** on the same tasks, graded by the same grader. Paid, so it runs only when you trigger it: Actions > Live evals > Run workflow, with the model ID as an input.
- **Live metrics:** pass rate, tokens, cost per task (from prices you pass), model and task latency p50/p99, turn-limit hits, and each final answer (shown, never graded).

HOW IT WORKS

Piece	Role
agents_openai.py	Chat Completions tool-calling loop. The model sees exactly the tools the server advertises, and every call goes through the harness's recorded call(), so it cannot grade itself.
ci.yml: evals	Replay agent + negative controls on every push and PR. Mark it a required check to block merges.
live-evals.yml	workflow_dispatch only, read-only token, key from the OPENAI_API_KEY secret, inputs passed via env (no script injection), 30-minute timeout, fails clearly if the secret is missing.

WHY IT WAS DONE THIS WAY

- **Free gate, paid measurement.** Replay is deterministic, so it can block merges. A model's score varies run to run, so live runs report a number and do not fail the build unless you ask (`--min-pass-rate`).
- **Manual trigger only.** No schedule and no PR trigger, so cost stays under your control and fork PRs can never reach the key.
- **Optional dependency.** openai is a separate install extra imported lazily; the server and the CI gate never install it. Tests use a fake client, so they need no key or network.

WHAT WENT WRONG OR SURPRISED ME (LOGGED IN FAILURES.MD)

- **Part 4's evals never tested what a model reads.** The replay agent ignores tool descriptions, so a misleading description passed every eval. The harness now hands each agent the server's own tool list; the live agent is scored on it.
- **One stray import would have broken CI.** CI does not install openai. I ran the full suite in a Python 3.12 venv without it (167 passed) and mutation-checked the agent loop: 5 of 5 mutations caught.

VERIFY IT YOURSELF

```
python -m evals.run       # 31/31 tasks passed; 12/12 negative controls caught
ruff check . && python -m pytest -q   # 167 passed (8 new: fake-client agent loop)
PR #6 > Checks > evals > Summary      # the report CI just produced
```

INTERVIEW TALKING POINT

"Every PR runs a free, deterministic trajectory eval that blocks merges. Real-model runs are a separate, owner-triggered measurement of pass rate, cost and latency, graded by the same code, and the API key is unreachable from pull requests."

NEXT

Your actions: mark **evals** a required check; add the OPENAI_API_KEY secret (and an OpenAI spending limit); trigger Live evals with your model ID. I then turn the first live run into the README's results table. Part 6 waits for your "go".